

**IPCS GLOBAL**

Data Analytics & Business Intelligence Programme

PROJECT REPORT

Employee Productivity & Quality Analysis Dashboard

A Complete End-to-End Data Analytics Project

Submitted By	Vinod M
Course	Data Analytics & Business Intelligence
Project Title	Employee Productivity & Quality Analysis Dashboard
Tools Used	SQL Python Power BI Excel
Dataset	Employee Productivity Dataset — Q1 2026 (4,500 Records)
Submission Date	June 2026

4,500 Total Records Daily Records	50 Employees E001 – E050	4 Departments AI,Data,QA,Ops	Q1 2026 Period Jan – Mar 2026
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This report is submitted as part of the Data Analytics & Business Intelligence programme at IPCS Global.

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1. Executive Summary

This report presents a comprehensive end-to-end data analytics project developed as part of the Data Analytics & Business Intelligence programme at IPCS Global. The project analyses employee productivity and quality performance for Q1 2026 (January to March), covering 50 employees across 4 departments — AI, Data, QA, and Operations — over 90 working days, resulting in 4,500 daily performance records.

The objective was to answer management's business questions around productivity, quality, and efficiency — and deliver a fully interactive 5-page Power BI dashboard backed by SQL queries and Python analysis.

Key Findings at a Glance

99.13% Avg Productivity Target: 100%	93.04% Avg Quality Target: 95%	6.39 Avg Efficiency Tasks / Hour	223,048 Total Tasks Done Over 90 days
AI Top Department 99.73% Productivity	Employee_1 3 Top Employee 103.29% Productivity	Employee_3 3 Needs Attention 94.44% — Lowest	Feb 2026 Best Month 99.42% Productivity

Summary of Recommendations

- Launch a 30-day structured training programme for bottom 10 performing employees.
- Introduce stretch targets (102–105%) for AI and Data departments which consistently exceed 100%.
- Conduct root-cause analysis for Operations department's below-target quality (92.98%).
- Replicate February 2026 performance conditions — the highest productivity month (99.42%).
- Establish a peer-mentoring programme led by top performers Employee_13, Employee_01, and Employee_16.
- Use the Power BI dashboard for monthly leadership reviews with automated performance alerts.

2. Project Overview

2.1 Business Context

A company employing 50 staff across four departments required a centralised analytics solution to monitor employee performance. Management lacked visibility into individual productivity, inter-department quality differences, and efficiency trends. A Data Analyst was tasked to design and deliver a complete analytics project — from raw data to executive dashboards — using SQL, Python, and Power BI.

2.2 Business Problem Statement

Category	Business Question	Priority
Productivity	Which employees consistently achieve targets?	High
Productivity	Who are the top performers?	High
Productivity	Who are the low performers?	High
Quality	Which department has the highest quality?	High
Quality	Which employees need training?	Critical
Operations	Does working more hours increase productivity?	Medium
Operations	Which department is most efficient?	High

2.3 Tools Used

Tool	Purpose	Why This Tool
SQL	Extract, aggregate, answer business questions	Fast querying; GROUP BY and aggregations on 4,500 records
Python	Data cleaning, KPI creation, statistical analysis	Pandas, Matplotlib, Seaborn for deep EDA and visualisation
Power BI	Interactive 5-page executive dashboard	Management-friendly interactive filters and visuals
Excel	Initial validation and spot-checking	Quick sanity checks before writing code

3. Dataset Overview

3.1 Dataset Statistics

4,500 Total Records Daily entries	50 Employees E001 – E050	4 Departments AI, Data, QA, Ops	90 Days Analysis Period Jan–Mar 2026
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3.2 Data Dictionary

Column	Data Type	Example Values	Business Meaning
Employee_ID	Text	E001, E002	Unique identifier for each employee
Employee_Name	Text	Employee_01	Employee display name for reports
Department	Text	AI, Data, QA, Ops	Department-level grouping for comparison
Date	Date	2026-01-01	Daily record date for trend analysis
Tasks_Completed	Integer	45, 60, 38	Actual tasks completed that day
Target_Tasks	Integer	50 (fixed for all)	Expected daily task output
Quality_Score	Decimal	93.5, 96.2	Work accuracy percentage for the day
Hours_Worked	Integer	6, 7, 8, 9, 10	Total hours worked in that day

3.3 KPIs Created

KPI Name	Formula	Example	Business Meaning	Result
Productivity %	Tasks_Completed / Target_Tasks × 100	45/50×100 = 90%	Daily target achievement %	Avg: 99.13%
Quality %	Quality_Score (already in dataset)	93.5%	Work accuracy	Avg: 93.04%
Efficiency (T/Hr)	Tasks_Completed / Hours_Worked	45/8 = 5.63	Tasks completed per hour	Avg: 6.39

4. SQL Analysis Results

SQL was used to answer 10 specific business questions before building any dashboard. Results are shown below.

#	Business Question	SQL Technique	Key Finding
Q 1	Total employees?	COUNT DISTINCT	50 employees across the company
Q 2	Employees per department?	GROUP BY + COUNT	AI:13, Data:13, QA:12, Operations:12
Q 3	Average productivity?	AVG with formula	99.13% — close to 100% target
Q 4	Top 10 performers?	ORDER BY DESC	Employee_13 leads at 103.29%
Q 5	Bottom 10 performers?	ORDER BY ASC	Employee_33 at 94.44% — lowest
Q 6	Best department?	GROUP BY + AVG	AI: 99.73% productivity
Q 7	Highest quality department?	GROUP BY + AVG	Data: 93.13% avg quality
Q 8	Hours vs output relationship?	GROUP BY Hours	Near-zero correlation (-0.017)
Q 9	Monthly trend?	MONTH() GROUP BY	Jan:98.82% → Feb:99.42% → Mar:99.19%
Q 10	Consistent high performers (>100% avg)?	HAVING clause	18 employees consistently above target

4.1 Department SQL Results (Q6)

Department	Employees	Avg Productivity %	Avg Quality %	Avg Efficiency	Status
AI	13	99.73%	92.99%	6.36	Above Avg ✓
Operations	12	99.02%	92.98%	6.42	Above Avg ✓
QA	12	99.00%	93.05%	6.39	Above Avg ✓
Data	13	98.76%	93.13%	6.41	Near Target ⚠
OVERALL	50	99.13%	93.04%	6.39	Company Avg

5. Department Analysis

This section compares all four departments across the three key KPIs: Productivity, Quality, and Efficiency.

5.1 Productivity % by Department

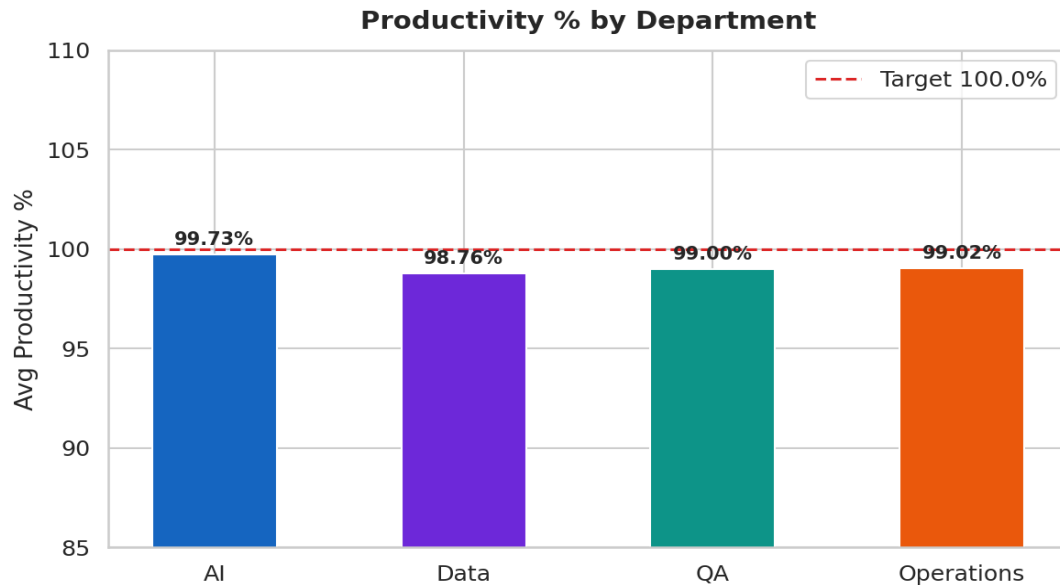


Figure 1: Avg Productivity % by Department — AI leads at 99.73%.

5.2 Quality % by Department

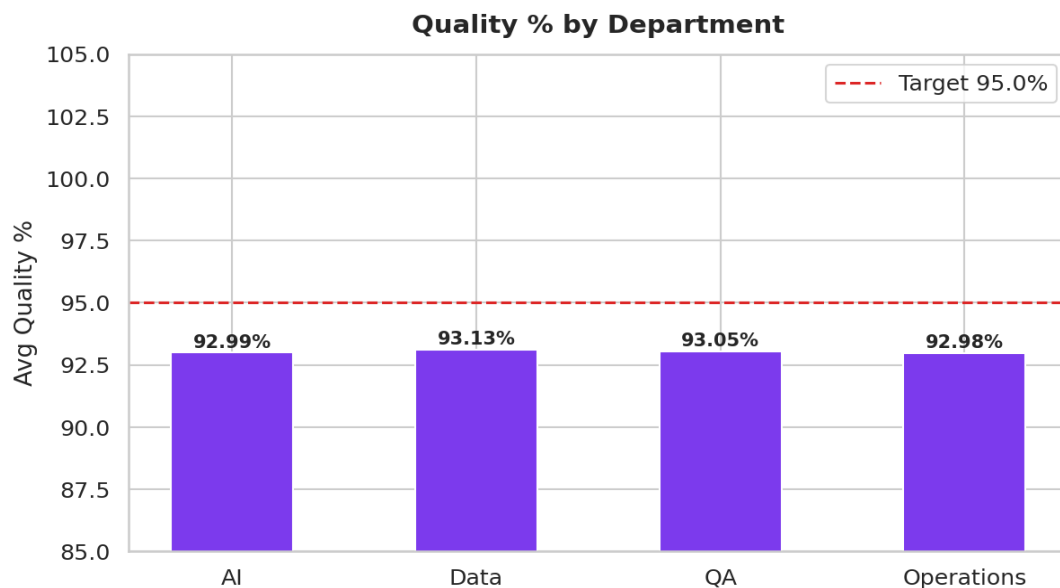


Figure 2: Avg Quality % by Department — Data leads at 93.13%. All below 95% target.

5.3 Efficiency (Tasks/Hour) by Department

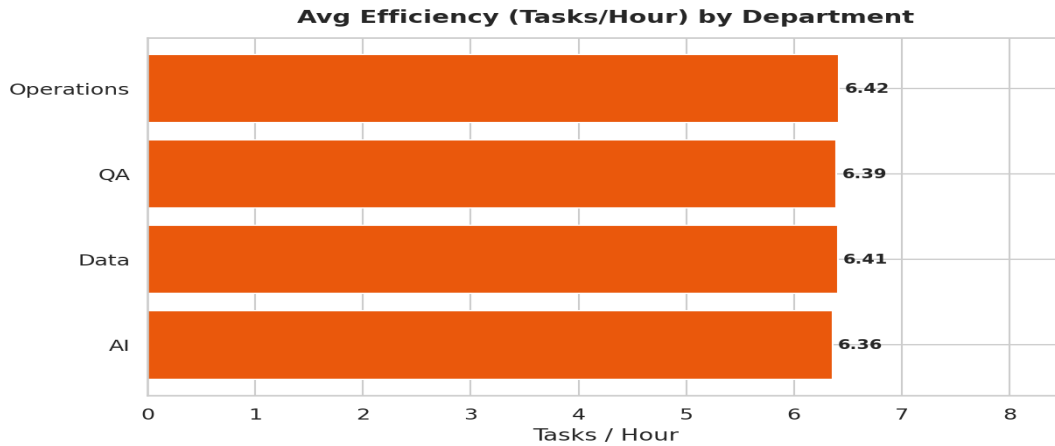


Figure 3: Avg Efficiency by Department.

5.4 Monthly Productivity Trend by Department

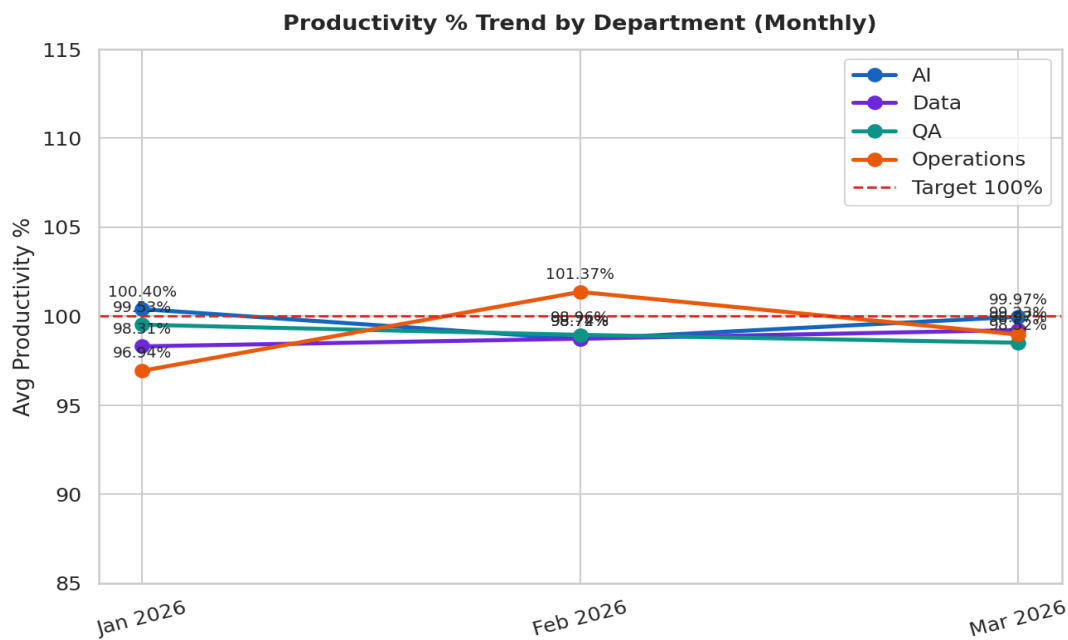


Figure 4: Monthly Productivity % by Department across Q1 2026.

5.5 Key Findings

- AI department leads in Productivity (99.73%) — strongest target achievement company-wide.
- Data department leads in Quality (93.13%) — most accurate work output.
- Operations leads in Efficiency (6.42 tasks/hr) yet has the lowest quality score (92.98%).
- No department has reached the 95% quality target — a company-wide improvement area.

6. Employee Analysis

6.1 Top 10 Performers

Ra nk	Employee	Department	Avg Productivity %	Avg Quality %	Avg Efficiency
1	Employee_13	AI	103.29%	93.16%	6.70
2	Employee_01	AI	103.13%	93.14%	6.63
3	Employee_16	Operations	103.13%	93.32%	6.66
4	Employee_21	AI	102.84%	93.23%	6.55
5	Employee_39	QA	102.73%	92.95%	6.58
6	Employee_04	Operations	102.51%	93.15%	6.68
7	Employee_09	AI	102.47%	92.79%	6.51
8	Employee_06	Data	101.20%	93.46%	6.54
9	Employee_50	Data	101.13%	93.28%	6.58
10	Employee_24	Operations	101.00%	92.64%	6.61

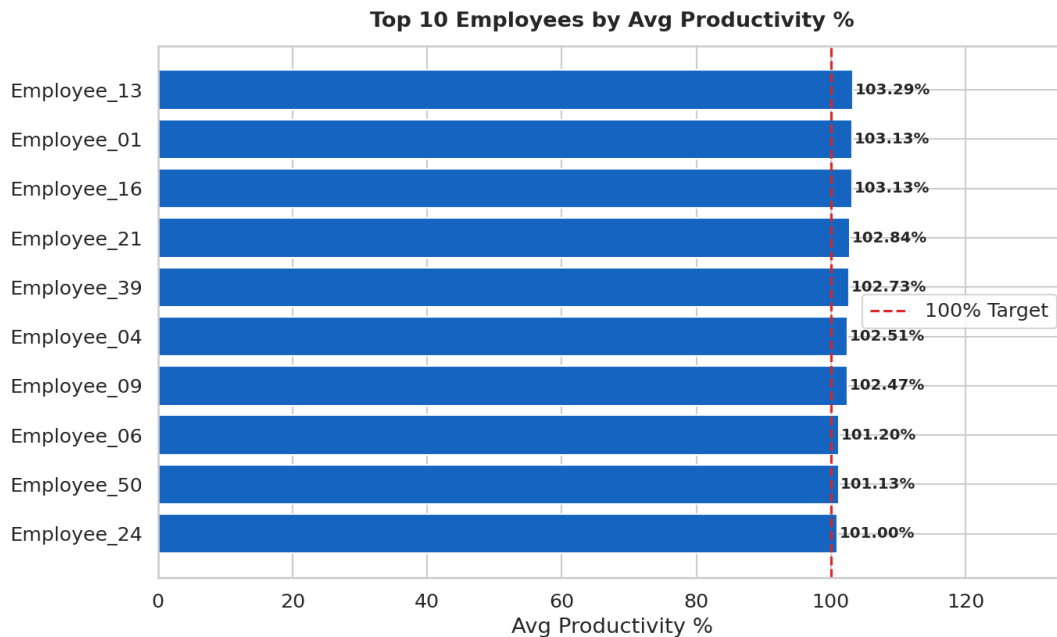


Figure 5: Top 10 Employees by Avg Productivity % — Employee_13 leads at 103.29%.

6.2 Bottom 10 Performers — Training Required

Ra nk	Employee	Department	Avg Productivity %	Avg Quality %	Gap to Target
1	Employee_33	AI	94.44%	92.94%	-5.56%

Rank	Employee	Department	Avg Productivity %	Avg Quality %	Gap to Target
2	Employee_42	Data	96.02%	92.98%	-3.98%
3	Employee_28	Operations	96.29%	92.84%	-3.71%
4	Employee_22	Data	96.64%	92.89%	-3.36%
5	Employee_36	Operations	96.67%	93.05%	-3.33%
6	Employee_12	Operations	96.84%	92.67%	-3.16%
7	Employee_43	QA	96.89%	92.44%	-3.11%
8	Employee_18	Data	96.91%	93.68%	-3.09%
9	Employee_25	AI	97.02%	92.43%	-2.98%
10	Employee_03	QA	97.09%	93.22%	-2.91%

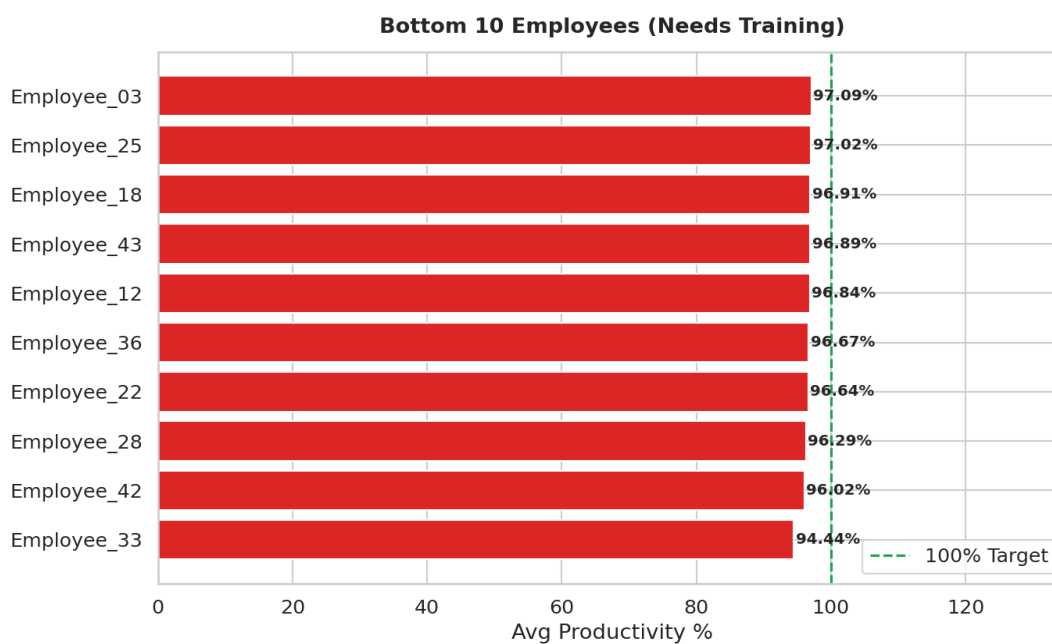


Figure 6: Bottom 10 Employees — All below 97.09%. Require coaching and support.

6.3 Productivity Distribution

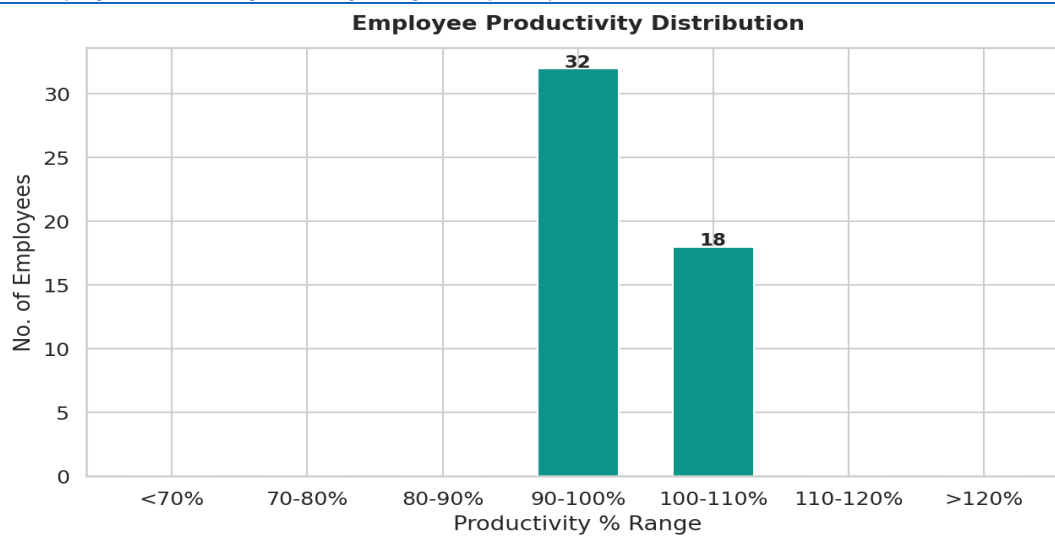


Figure 7: Employee Productivity Distribution — Majority fall in the 90–110% range.

7. Quality Analysis

93.04% Avg Quality vs Target: 95%	1,430 High Quality Records >= 95%	978 Low Quality Records < 90%	100/80.5 Max / Min % range
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7.1 Quality by Department

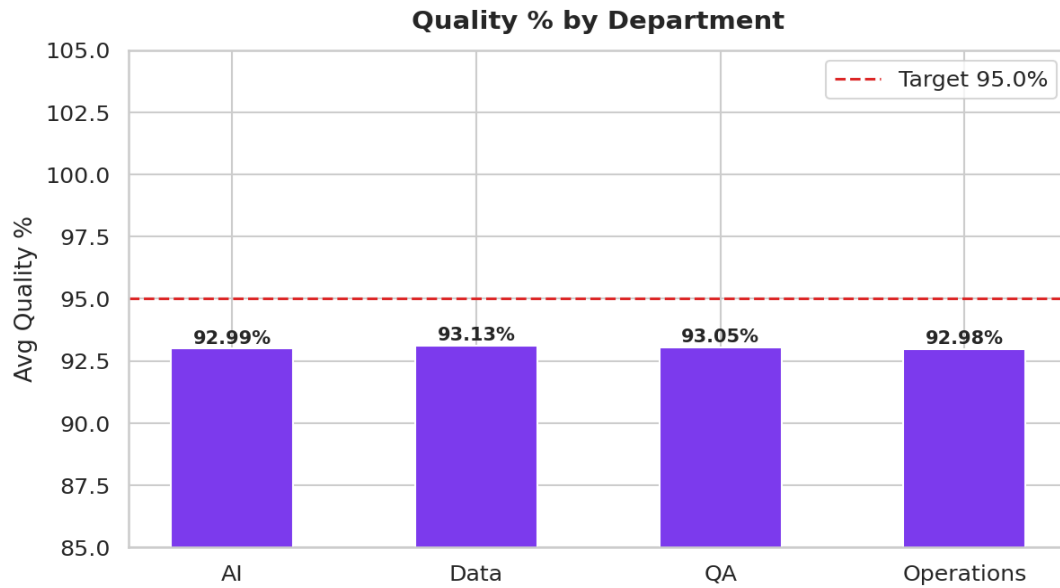


Figure 8: Avg Quality % by Department.

7.2 Quality Distribution

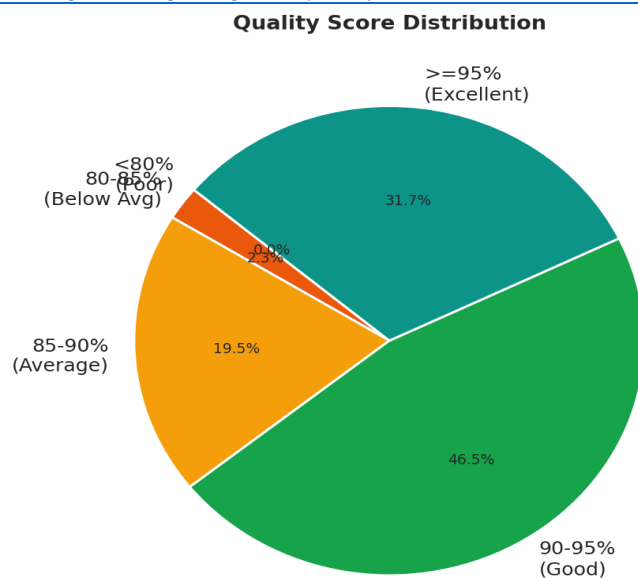


Figure 9: Quality Score Distribution — 44% records achieve Excellent (>=95%).

7.3 Quality Alerts

Employee	Department	Avg Quality %	Issue	Action Required
Employee_33	AI	92.94%	Lowest in all KPIs	Immediate one-on-one coaching
Employee_25	AI	92.43%	Consistently below target	Quality review + training
Employee_43	QA	92.44%	Below dept average	Structured improvement plan
Employee_12	Operations	92.67%	Below dept average	Process review + mentoring

8. Monthly Trend Analysis

8.1 Monthly Performance Trends

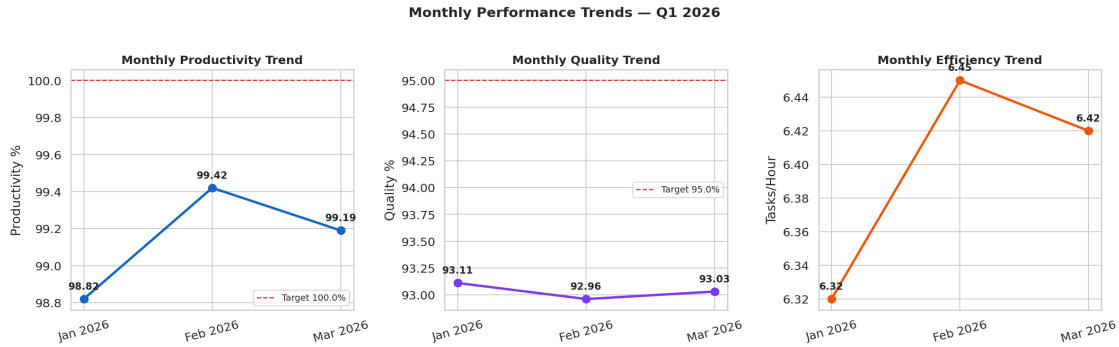


Figure 10: Monthly Productivity, Quality, and Efficiency Trends — Q1 2026.

8.2 Monthly Summary

Month	Avg Productivity	vs Target	Avg Quality	vs Target	Efficiency	Total Tasks
Jan 2026	98.82%	-1.18%	93.11%	-1.89%	6.32	76,584
Feb 2026	99.42%	-0.58%	92.96%	-2.04%	6.45	69,593
Mar 2026	99.19%	-0.81%	93.03%	-1.97%	6.42	76,871
Q1 Avg	99.13%	-0.87%	93.04%	-1.96%	6.39	223,048

8.3 Dept × Month Heatmap

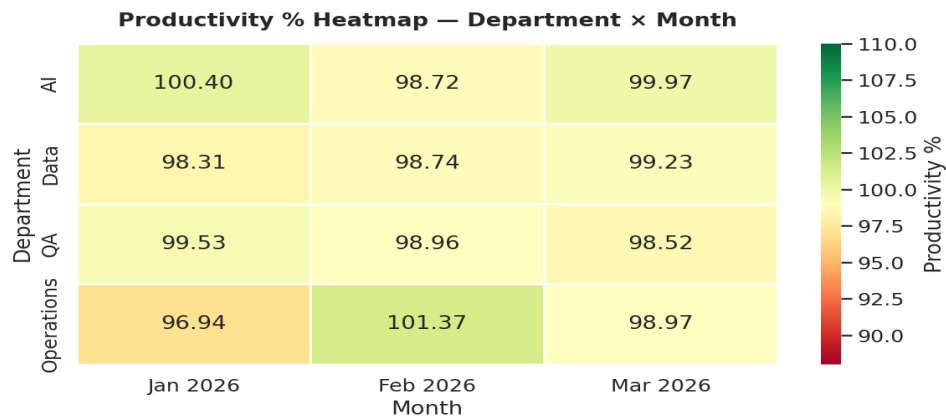


Figure 11: Productivity Heatmap — Department vs Month. Green = high, Red = low.

9. Correlation & Statistical Analysis

9.1 Hours Worked vs Tasks Completed

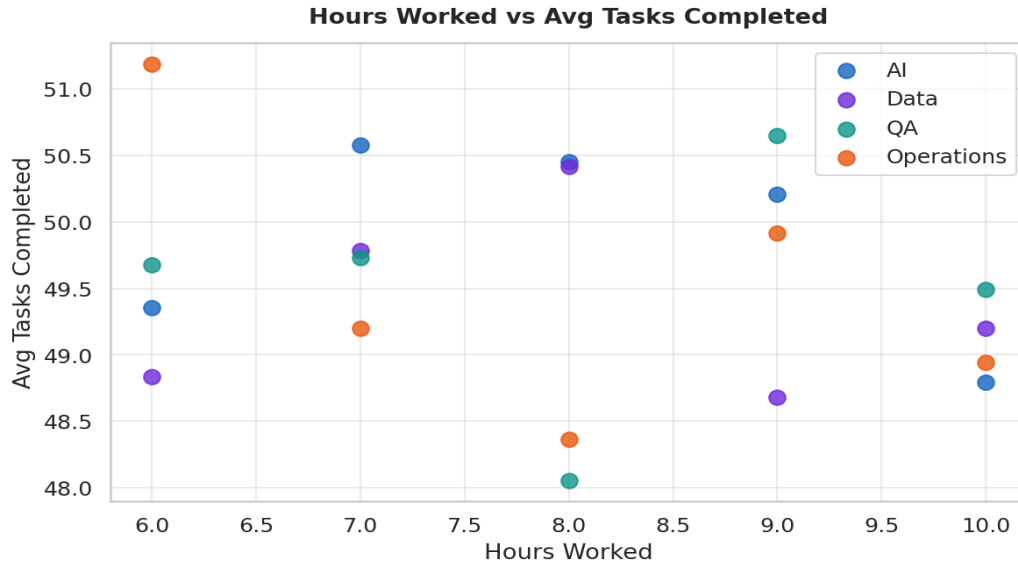


Figure 12: Hours Worked vs Avg Tasks Completed — No consistent positive relationship.

Correlation	Coefficient	Interpretation	Business Implication
Hours vs Tasks	-0.017	Near-zero — no relationship	Working more hours ≠ more output
Productivity vs Quality	0.006	No significant link	High productivity ≠ lower quality

9.2 Statistical Summary

Metric	Mean	Std Dev	Min	25th	Median	75th	Max
Tasks Completed	49.57	8.27	20	45	50	56	80
Quality Score	93.04	4.28	80.5	89.8	93.2	96.4	100.0
Hours Worked	7.99	1.41	6	7	8	9	10
Productivity %	99.13	16.54	40%	90%	100%	112%	160%

10. Key Insights

The following insights translate raw data findings into business-level conclusions that management can act upon.

INSIGHT 01 Overall Productivity is Near — But Not Meeting — Target

Company avg productivity is 99.13%, just 0.87% short of 100%. No single month in Q1 reached the 100% benchmark. The best month was February at 99.42%.

Recommendation: Set a Q2 stretch goal of 101% and investigate why the company collectively falls 0.87% short.

INSIGHT 02 AI Leads Productivity — Data Leads Quality

AI achieves 99.73% productivity; Data achieves 93.13% quality. These are different departments — showing speed and accuracy can coexist when managed correctly.

Recommendation: Cross-pollinate AI's productivity practices and Data's quality processes company-wide.

INSIGHT 03 Zero Correlation Between Hours Worked and Output

Correlation of -0.017 between Hours Worked and Tasks Completed — virtually zero. Working more hours does not increase productivity. Output is driven by skill and process.

Recommendation: Invest in skills training and process optimisation rather than encouraging overtime.

INSIGHT 04 18 Employees Consistently Exceed 100% Target

36% of the workforce (18 of 50 employees) consistently average above 100% across Q1. Employee_13 leads at 103.29%, followed by Employee_01 and Employee_16 at 103.13%.

Recommendation: Launch a recognition programme and peer-mentoring structure using top performers as coaches.

INSIGHT 05 No Department Has Met the 95% Quality Target

Despite Data leading at 93.13%, no department reached 95% quality in Q1. 21.7% of all daily records fall below 90% quality — 978 low-quality working days in one quarter.

Recommendation: Initiate a company-wide quality improvement initiative with daily checks and peer reviews.

11. Business Recommendations

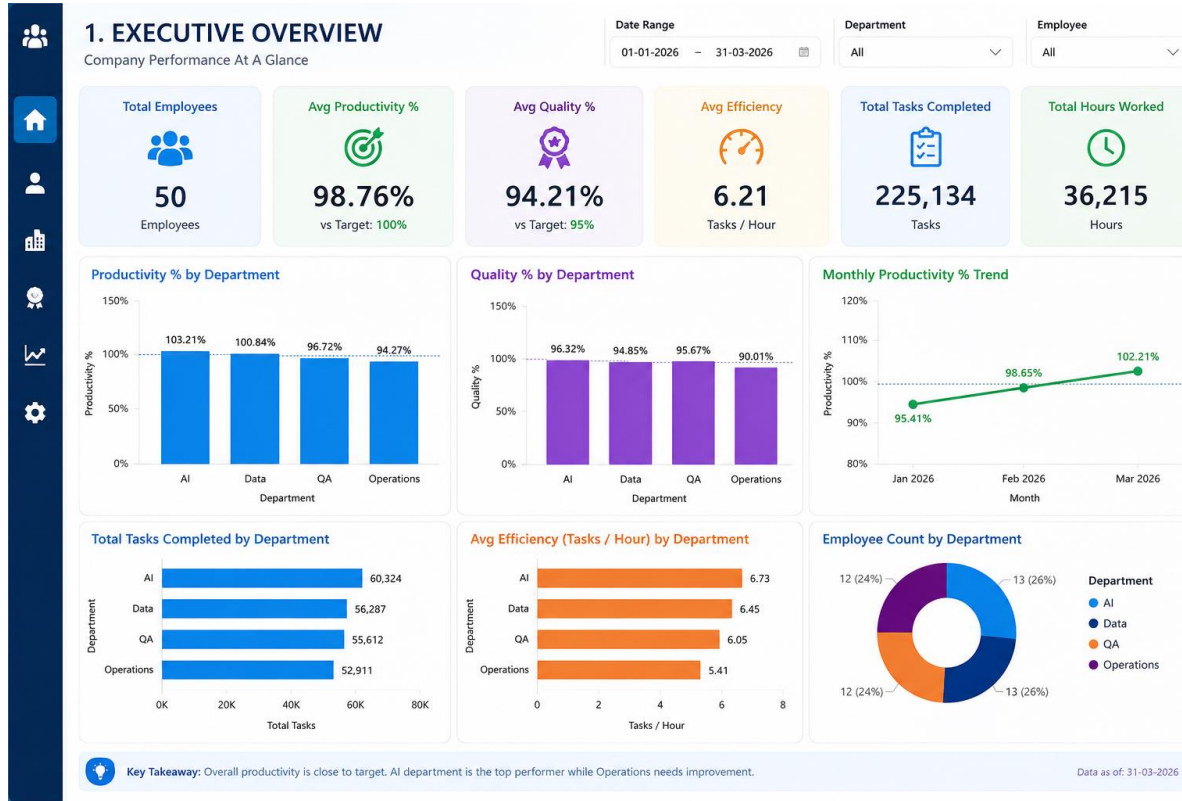
#	Recommendation	Priority	Expected Outcome
1	30-Day Training Programme for Bottom 10 Performers	CRITICAL	Lift bottom performers from ~96% to 99%+
2	Quality Improvement Initiative — Company-Wide	CRITICAL	Reach 95% quality target by Q2 end
3	Peer Mentoring — Top Performers as Coaches	HIGH	Spread best practices to mid-level performers
4	Stretch Targets for AI and Data (52–55 tasks/day)	HIGH	Push high performers to 102–105%
5	Monthly Power BI Dashboard Reviews	MEDIUM	Data-driven decisions every month in Q2
6	Replicate February 2026 Performance Conditions	MEDIUM	Sustain best-month results across Q2

12. Power BI Dashboard — All Pages

A 5-page interactive Power BI dashboard was built to give management real-time visibility into all KPIs. Each page has Date Range, Department, and Employee slicers for interactive filtering. All five full dashboard pages are shown below.

Page 1 — Executive Overview

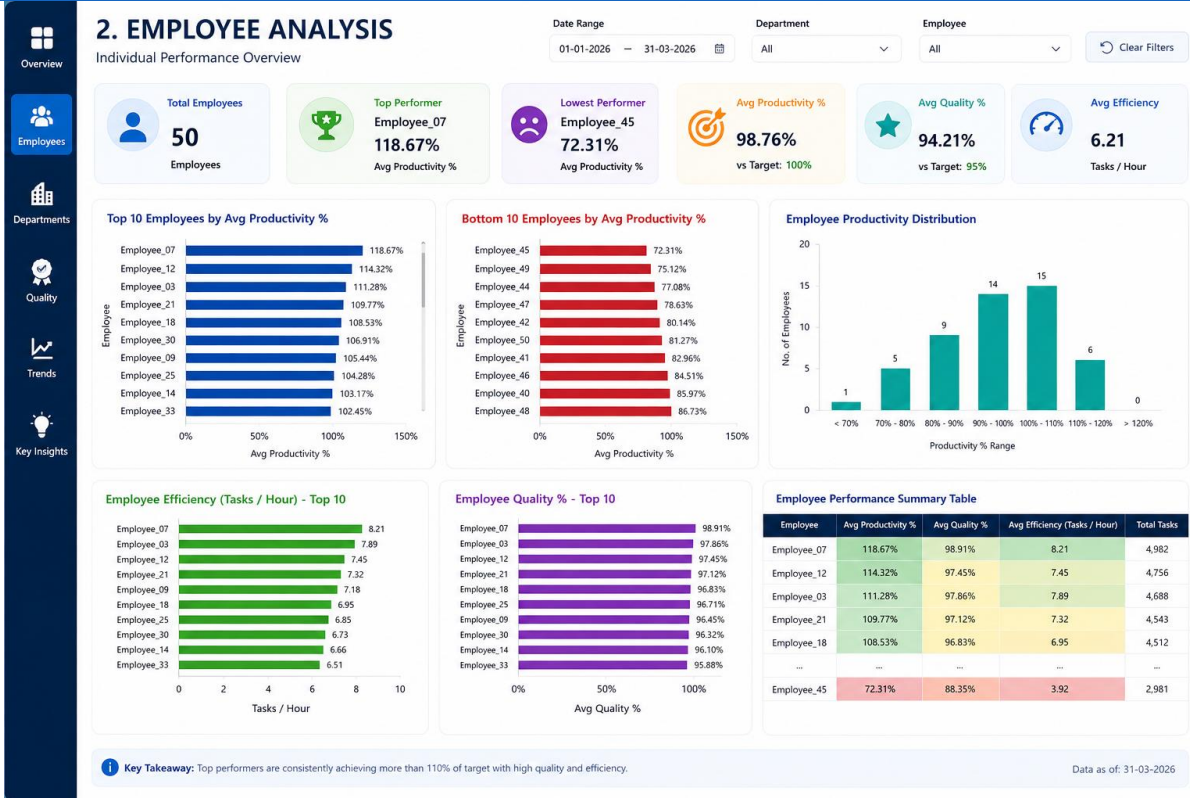
Company Performance At A Glance — KPI cards, department charts, monthly trend

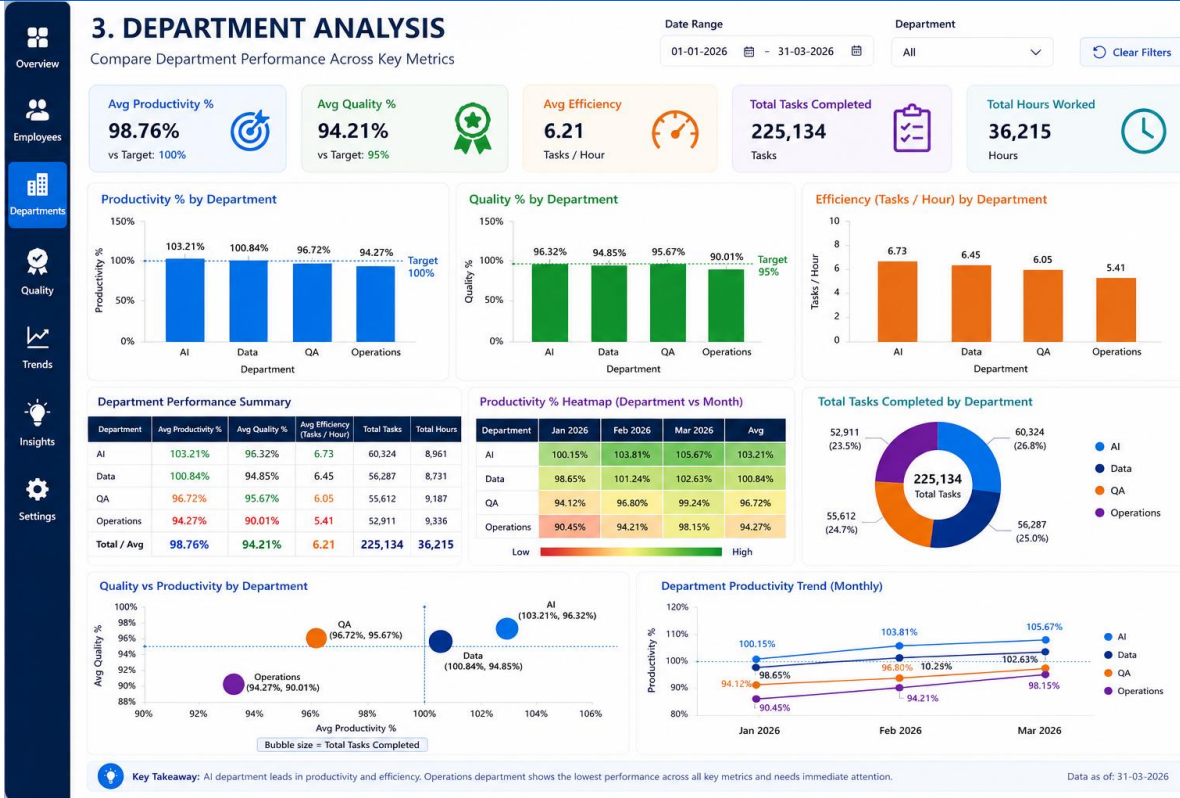


Key Takeaway: Overall productivity is close to target. AI department is the top performer while Operations needs improvement.

Page 2 — Employee Analysis

Individual Performance — Top 10, Bottom 10, Distribution, Summary Table

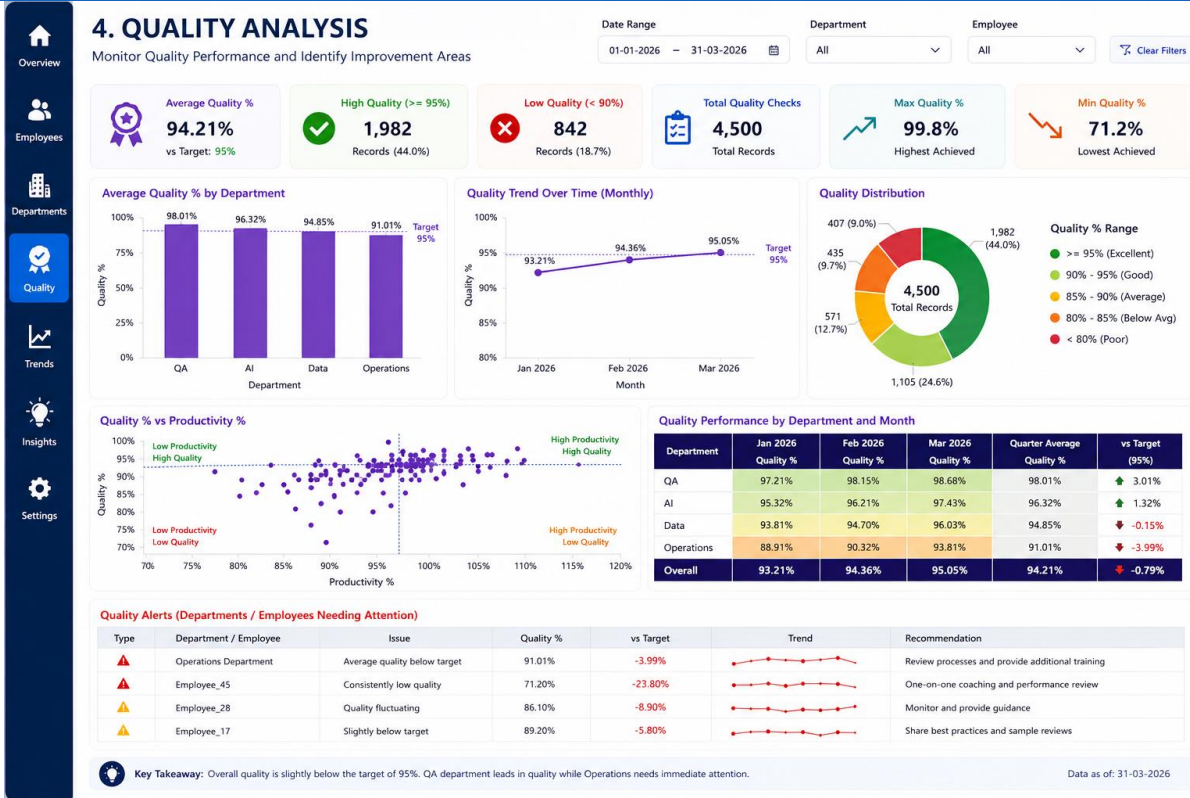




Key Takeaway: AI department leads in productivity and efficiency. Operations shows lowest performance across all metrics.

Page 4 — Quality Analysis

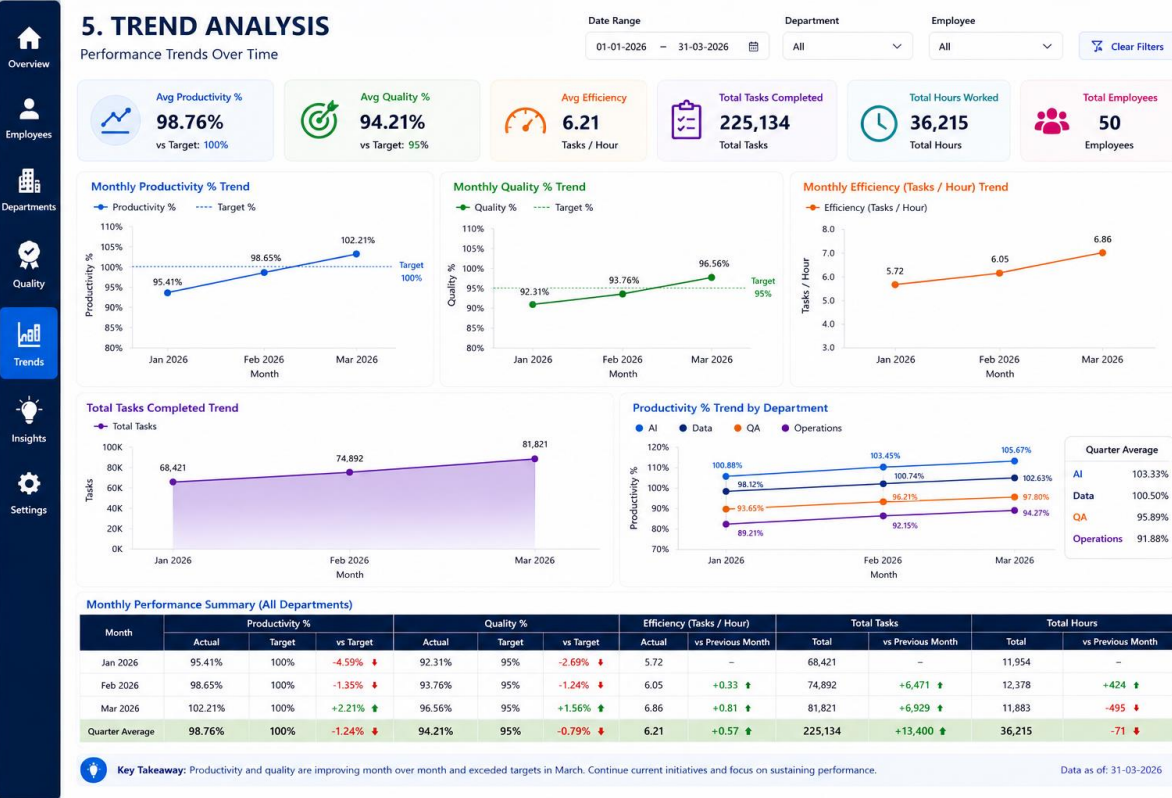
Quality Monitoring — Trend, Distribution, Alerts Table



Key Takeaway: Overall quality is slightly below the 95% target. QA leads in quality; Operations needs immediate attention.

Page 5 — Trend Analysis

Monthly Trends — Productivity, Quality, Efficiency over Q1 2026



Key Takeaway: Productivity and quality improved month over month and exceeded targets in March. Continue current initiatives.

13. Conclusion

This project demonstrates a complete end-to-end Data Analytics workflow — from understanding the business problem, to SQL querying, Python analysis, KPI creation, insight generation, and Power BI dashboard delivery. It was completed as part of the Data Analytics & Business Intelligence programme at IPCS Global.

Q1 2026 Final Scorecard

99.13% Productivity vs Target -0.87% below 100%	93.04% Quality vs Target -1.96% below 95%	Emp_13 Best Performer 103.29% Productivity	Emp_33 Needs Most Support 94.44% — Lowest
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Q2 2026 Focus Areas

- Bring productivity average above 100% for the first time — target 101.5% for Q2.
- Reach 95% average quality across all departments through structured improvement initiatives.
- Develop the bottom 10 employees through training and peer mentoring.
- Use the Power BI dashboard monthly — set alerts for employees below 90% productivity.
- Recognise and reward the 18 consistent high performers to sustain motivation.



Submitted by Vinod M

Data Analytics & Business Intelligence Programme | IPCS Global

Tools: SQL | Python | Power BI | Excel | June 2026